📘 MSU-IIT SASE 2025 Mathematics Reviewer

Flashcards / Trivia / General Knowledge

🔢 Arithmetic & Number Theory (Q1–Q50)

Q1. What is the smallest prime number?
A1. 2

Q2. What is the largest two-digit prime number?
A1. 97

Q3. What is the LCM of 18 and 24?
A1. 72

Q4. What is the GCF of 144 and 216?
A1. 72

Q5. Express 0.75 as a fraction in simplest form.
A1. $\frac{3}{4}$

Q6. Write 3500 in scientific notation.
A1. 3.5×10^3

Q7. Round 4576 to the nearest hundred.
A1. 4600

Q8. What is 15×16?
A1. 240

Q9. What is the remainder of 205 ÷ 6?
A1. 1

Q10. What is 12% of 750?
A1. 90

Q11. If a car travels 180 km in 3 hours, what is its average speed?
A1. 60 km/h

Q12. If 5 men can finish a job in 10 days, how many days will 10 men take?
A1. 5 days

Q13. The ratio of boys to girls is 3:5. If there are 24 girls, how many boys?
A1. 15

Q14. Convert 0.2 into a percentage.
A1. 20%

Q15. What is the simple interest of ₱5000 at 6% in 2 years?
A1. ₱600

Q16. Solve for x: $12x = 96$.
A1. $x = 8$

Q17. The sum of the first 10 natural numbers?
A1. 55

Q18. The sum of the first 100 positive integers?
A1. 5050

Q19. The square root of 484 is?
A1. 22

Q20. What is 25^2?
A1. 625

Q21. What is 2^5?
A1. 32

Q22. What is the cube root of 729?
A1. 9

Q23. What is 20% of 360?
A1. 72

Q24. What is 15% of ₱2000?
A1. ₱300

Q25. Simplify: $\frac{2}{3} \times \frac{9}{10}$.
A1. $\frac{3}{5}$

Q26. Simplify: $\frac{7}{8} \div \frac{21}{32}$.
A1. $\frac{4}{3}$

Q27. Write in decimal: $\frac{7}{20}$.
A1. 0.35

Q28. Find the arithmetic mean of 6, 8, 12, 14.
A1. 10

Q29. Convert 5 km to meters.
A1. 5000 m

Q30. Convert 2.5 hours into minutes.
A1. 150 minutes

Q31. Express 3/8 as a percentage.
A1. 37.5%

Q32. What is the perimeter of a square with side 15 cm?
A1. 60 cm

Q33. If 12 mangoes cost ₱96, what is the cost of 20 mangoes?
A1. ₱160

Q34. Which is divisible by 11: 253, 308, 462?
A1. 308

Q35. What is the next prime after 89?

A1. 97

Q36. How many prime numbers are less than 20?

A1. 8 (2, 3, 5, 7, 11, 13, 17, 19)

Q37. Which is a perfect square: 128, 144, 150?

A1. 144

Q38. Which is a perfect cube: 216, 234, 250?

A1. 216

Q39. Find the least common multiple of 5, 6, and 10.

A1. 30

Q40. Find the greatest common divisor of 36 and 81.

A1. 9

Q41. Solve: 3^4.

A1. 81

Q42. Write 0.001 as a power of 10.

A1. 10^{-3}

Q43. Find the value: $\sqrt{196}$.

A1. 14

Q44. If the selling price is ₱120 and the cost is ₱100, what is the profit percent?

A1. 20%

Q45. What is the value of $\frac{1}{0}$?

A1. Undefined

Q46. Express 1.25 as a fraction in lowest terms.

A1. $\frac{5}{4}$

Q47. Find the next term: 2, 4, 8, 16, ...

A1. 32

Q48. What is the digit sum of 98765?

A1. 35

Q49. Find the average of first five odd numbers.

A1. 5

Q50. Which is irrational: $\sqrt{36}$, $\sqrt{50}$, $\sqrt{81}$?

A1. $\sqrt{50}$

📊 Algebra & Functions (Q51–Q100)

Q51. Solve for x: $2x + 5 = 15$.
A1. $x = 5$

Q52. Solve: $3x - 7 = 11$.
A1. $x = 6$

Q53. Factorize: $x^2 - 9$.
A1. $(x - 3)(x + 3)$

Q54. Expand: $(x + 3)^2$.
A1. $x^2 + 6x + 9$

Q55. Expand: $(2x - 1)(x + 4)$.
A1. $2x^2 + 7x - 4$

Q56. Factorize: $x^2 - 5x + 6$.
A1. $(x - 2)(x - 3)$

Q57. Factorize: $x^2 + 7x + 12$.
A1. $(x + 3)(x + 4)$

Q58. Solve: $x^2 - 5x + 6 = 0$.
A1. $x = 2, 3$
Q59. Solve: $x^2 - 4x - 21 = 0$.
A1. $x = 7, -3$

Q60. Solve: $2x^2 - 8x + 6 = 0$.
A1. $x = 1, 3$

Q61. Slope of line $y = 3x + 2$.
A1. 3

Q62. Slope of line $2x + 3y = 6$.
A1. $-\frac{2}{3}$

Q63. Intercepts of $y = -2x + 4$.
A1. y-intercept 4, x-intercept 2

Q64. Vertex of parabola $y = x^2 - 4x + 3$.
A1. (2, -1)

Q65. Axis of symmetry of $y = x^2 - 6x + 8$.
A1. $x = 3$

Q66. Solve: $\frac{5}{x} = 2$.
A1. $x = \frac{5}{2}$

Q67. Evaluate $f(x) = 3x + 1$ at $x = 4$.
A1. 13

Q68. If $f(x) = x^2 - 1$, find $f(5)$.
A1. 24

Q69. If $f(x) = 2x + 3$, what is $f^{-1}(x)$?
A1. $\frac{x-3}{2}$

Q70. What is the discriminant of $2x^2 - 4x + 1 = 0$?
A1. 8

Q71. Solve: $|x - 3| = 5$.
A1. $x = 8$ or $x = -2$

Q72. Solve: $|2x - 7| = 3$.
A1. $x = 5$ or $x = 2$

Q73. Solve: $2(x + 3) = 4x - 6$.
A1. $x = 6$

Q74. Solve: $3(x - 2) + 4 = 2x$.
A1. $x = 10$

Q75. Solve for x: $\frac{x}{4} + \frac{x}{6} = 5$.
A1. $x = 12$

Q76. If $y = 2x^2 - 8$, find y when $x = -3$.
A1. 10

Q77. If $f(x) = \sqrt{x}$, evaluate $f(25)$.
A1. 5

Q78. What are the roots of $(x - 5)(x + 2) = 0$?
A1. $x = 5, -2$

Q79. Solve: $(x - 3)^2 = 16$.
A1. $x = 7, -1$

Q80. Solve: $(x + 4)(x - 2) = 0$.
A1. $x = -4, 2$

Q81. Simplify: $(2x^3)(3x^2)$.
A1. $6x^5$

Q82. Simplify: $(x^2)^3$.
A1. x^6

Q83. Simplify: $x^a \cdot x^b$.
A1. x^{a+b}

Q84. Simplify: $\frac{x^7}{x^3}$.
A1. x^4

Q85. What is x^0 for $x \neq 0$?
A1. 1

Q86. Solve: $9^x = 81$.
A1. $x = 2$

Q87. Solve: $2^{x+1} = 16$.
A1. $x = 3$

Q88. Simplify: $\sqrt{50}$.
A1. $5\sqrt{2}$

Q89. Simplify: $\sqrt{72}$.
A1. $6\sqrt{2}$

Q90. Solve: $\sqrt{x+1} = 5$.
A1. $x = 24$

Q91. Solve: $\sqrt{3x} = 9$.
A1. $x = 27$

Q92. Simplify: $(a+b)^2 - (a-b)^2$.
A1. 4ab

Q93. Factorize: $x^2 + 2x + 1$.
A1. $(x+1)^2$

Q94. Factorize: $x^2 - 10x + 25$.
A1. $(x-5)^2$
Q95. If $y = 2x^2 - 5x + 3$, find y when $x = 2$.
A1. 1

Q96. Solve: $5^x = 125$.
A1. $x = 3$

Q97. Solve: $\log_2 8 = x$.
A1. $x = 3$

Q98. Simplify: $\log 1000$.
A1. 3

Q99. What is $\ln e^5$?
A1. 5

Q100. What is $\log_{10} 1$?
A1. 0

📐 Geometry (Q101–Q150)

Q101. What is the sum of interior angles of a triangle?
A101. 180°

Q102. What is the sum of interior angles of a hexagon?
A102. 720°

Q103. What is the sum of interior angles of a 12-gon?
A103. 1800°

Q104. Each interior angle of a regular pentagon is?
A104. 108°

Q105. Each interior angle of a regular octagon is?
A105. 135°

Q106. The diagonal of a square is 10. Find its side.
A106. $5\sqrt{2}$

Q107. The area of an equilateral triangle with side 6.
A107. $9\sqrt{3}$

Q108. Perimeter of a circle is also called?
A108. Circumference

Q109. Area of a circle with radius 7?
A109. 49π

Q110. Diameter of a circle is 10. Find radius.
A110. 5

Q111. The volume of a cube with edge 5.
A111. 125

Q112. Surface area of a cube with edge 3.
A112. 54

Q113. Volume of a cylinder with radius 3 and height 10.
A113. 90π

Q114. Volume of a cone with radius 3 and height 4.
A114. 12π

Q115. Volume of a sphere with radius 6.
A115. 288π

Q116. The 3D shape with 6 rectangular faces.
A116. Rectangular prism (cuboid)

Q117. The 3D shape with 1 circular base and 1 vertex.
A117. Cone

Q118. The 3D shape with 2 circular bases.
A118. Cylinder

Q119. The number of faces of a cube.
A119. 6

Q120. The number of edges of a cube.
A120. 12

Q121. The number of vertices of a cube.
A121. 8

Q122. What is Euler's formula for polyhedra?
A122. $V - E + F = 2$

Q123. Area of rectangle with length 8 and width 5.
A123. 40

Q124. Perimeter of rectangle with length 12 and width 7.
A124. 38

Q125. Area of trapezoid with bases 10, 6 and height 4.
A125. 32

Q126. Area of parallelogram with base 8 and height 5.
A126. 40
Q127. Area of rhombus with diagonals 10 and 8.
A127. 40

Q128. Length of diagonal in right rectangle with sides 9 and 12.
A128. 15

Q129. Pythagorean theorem states?
A129. $a^2 + b^2 = c^2$

Q130. A triangle with angles 90°, 60°, and 30° is called?
A130. 30–60–90 right triangle

Q131. Hypotenuse of right triangle with legs 8 and 15.
A131. 17

Q132. Triangle with all sides equal is?
A132. Equilateral triangle

Q133. Triangle with two equal sides is?
A133. Isosceles triangle

Q134. Triangle with all sides unequal is?
A134. Scalene triangle

Q135. Circle with radius equal to 1 unit is called?
A135. Unit circle

Q136. Line touching a circle at exactly one point.
A136. Tangent

Q137. Line passing through two points of a circle.
A137. Chord (or secant if extended)

Q138. Longest chord of a circle.
A138. Diameter

Q139. Formula for distance between two points (x_1, y_1), (x_2, y_2).
A139. $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Q140. Midpoint of segment with ends (x_1, y_1), (x_2, y_2).
A140. $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

Q141. Equation of a circle with center (0,0) radius r.
A141. $x^2 + y^2 = r^2$

Q142. Equation of a line with slope m passing through (x_1, y_1).
A142. $y - y_1 = m(x - x_1)$

Q143. Equation of parabola with vertex at origin opening upward.
A143. $y = ax^2$

Q144. Area of sector formula?
A144. $\frac{\theta}{360} \cdot \pi r^2$

Q145. Arc length formula?
A145. $\frac{\theta}{360} \cdot 2\pi r$

Q146. If circumference = 44 cm, radius = ? ($\pi = 22/7$).
A146. 7 cm

Q147. Regular polygon with 8 sides is called?
A147. Octagon

Q148. Regular polygon with 12 sides?
A148. Dodecagon

Q149. Regular polygon with 20 sides?
A149. Icosagon

Q150. Number of diagonals in a decagon.
A150. 35

📐📏 Trigonometry (Q151–Q200)

Q151. Value of $\sin 0^\circ$.
A151. 0

Q152. Value of $\cos 0^\circ$.
A152. 1

Q153. Value of $\sin 30^\circ$.
A153. $\frac{1}{2}$

Q154. Value of $\cos 30^\circ$.
A154. $\frac{\sqrt{3}}{2}$

Q155. Value of $\tan 30^\circ$.
A155. $\frac{1}{\sqrt{3}}$

Q156. Value of $\sin 45^\circ$.
A156. $\frac{\sqrt{2}}{2}$

Q157. Value of $\cos 45^\circ$.
A157. $\frac{\sqrt{2}}{2}$

Q158. Value of $\tan 45^\circ$.
A158. 1

Q159. Value of $\sin 60^\circ$.
A159. $\frac{\sqrt{3}}{2}$

Q160. Value of $\cos 60^\circ$.
A160. $\frac{1}{2}$

Q161. Value of $\tan 60^\circ$.
A161. $\sqrt{3}$

Q162. Value of $\sin 90^\circ$.
A162. 1

Q163. Value of $\cos 90^\circ$.
A163. 0

Q164. Value of $\tan 90^\circ$.
A164. Undefined

Q165. Convert 180° to radians.
A165. π

Q166. Convert 270° to radians.
A166. $\frac{3\pi}{2}$

Q167. Convert 360° to radians.
A167. 2π

Q168. Period of sine and cosine functions.
A168. 2π

Q169. Period of tangent function.
A169. π

Q170. Reciprocal of sine.
A170. Cosecant (csc)

Q171. Reciprocal of cosine.

A171. Secant (sec)

Q172. Reciprocal of tangent.

A172. Cotangent (cot)

Q173. Pythagorean identity.

A173. $\sin^2\theta + \cos^2\theta = 1$

Q174. Double angle formula for sine.

A174. $\sin 2\theta = 2\sin\theta\cos\theta$

Q175. Double angle formula for cosine.

A175. $\cos 2\theta = \cos^2\theta - \sin^2\theta$

Q176. Double angle formula for tangent.

A176. $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$

Q177. If $\sin\theta = \frac{3}{5}$, find $\cos\theta$.

A177. $\frac{4}{5}$

Q178. If $\cos\theta = \frac{12}{13}$, find $\sin\theta$.

A178. $\frac{5}{13}$

Q179. $\sin^2 45^\circ + \cos^2 45^\circ$.

A179. 1

Q180. $\sin 90^\circ \cdot \cos 0^\circ$.

A180. 1

Q181. Angle in radians of 120°.

A181. $\frac{2\pi}{3}$

Q182. Angle in radians of 225°.

A182. $\frac{5\pi}{4}$

Q183. Angle in radians of 315°.

A183. $\frac{7\pi}{4}$

Q184. If $\tan A = \frac{3}{4}$, then $\sin A = ?$

A184. $\frac{3}{5}$

Q185. If $\tan A = \frac{3}{4}$, then $\cos A = ?$

A185. $\frac{4}{5}$

Q186. The law of sines states?

A186. $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Q187. The law of cosines states?

A187. $c^2 = a^2 + b^2 - 2ab\cos C$

Q188. Height of an equilateral triangle of side a.

A188. $\frac{\sqrt{3}}{2}a$

Q189. Area of a triangle given 2 sides and included angle.
A189. $\frac{1}{2}ab\sin C$

Q190. Area of triangle using Heron's formula.
A190. $\sqrt{s(s-a)(s-b)(s-c)}$

Q191. Quadrant where both sine and cosine are positive.
A191. First quadrant

Q192. Quadrant where sine is positive but cosine is negative.
A192. Second quadrant

Q193. Quadrant where tangent is positive but sine is negative.
A193. Third quadrant

Q194. Quadrant where cosine is positive but sine is negative.
A194. Fourth quadrant

Q195. $\sin(-\theta) = ?$
A195. $-\sin\theta$

Q196. $\cos(-\theta) = ?$
A196. $\cos\theta$

Q197. $\tan(-\theta) = ?$
A197. $-\tan\theta$

Q198. $\sin(180^\circ - \theta) = ?$
A198. $\sin\theta$

Q199. $\cos(180^\circ - \theta) = ?$
A199. $-\cos\theta$

Q200. $\tan(180^\circ + \theta) = ?$
A200. $\tan\theta$

📊 Statistics & Probability (Q201–Q250)

Q201. What is the mean of 5, 10, 15, 20?
A201. 12.5

Q202. What is the median of 5, 7, 9, 11, 13?
A202. 9

Q203. What is the mode of 2, 3, 3, 5, 6?
A203. 3

Q204. Find the range of 8, 4, 10, 15, 2.
A204. 13

Q205. Mean of first 10 natural numbers.
A205. 5.5

Q206. Population standard deviation symbol.
A206. σ

Q207. Sample standard deviation symbol.
A207. s

Q208. Variance formula in terms of mean.
A208. $\sigma^2 = \frac{\sum(x_i-\mu)^2}{N}$

Q209. What is the probability of getting heads in a fair coin toss?
A209. 0.5

Q210. Probability of rolling a 6 on a fair die?

A210. $\frac{1}{6}$

Q211. Probability of rolling an even number on a die?

A211. $\frac{3}{6} = \frac{1}{2}$

Q212. Probability of drawing a heart from a deck of cards.

A212. $\frac{13}{52} = \frac{1}{4}$

Q213. Probability of drawing an ace from a deck.

A213. $\frac{4}{52} = \frac{1}{13}$

Q214. Probability of drawing a king or queen.

A214. $\frac{8}{52} = \frac{2}{13}$

Q215. Complement rule in probability.

A215. $P(A^c) = 1 - P(A)$

Q216. Mutually exclusive events are?

A216. Events that cannot occur at the same time

Q217. Independent events are?

A217. Events where outcome of one does not affect the other

Q218. Probability of both independent events A and B occurring.

A218. $P(A \cap B) = P(A) \cdot P(B)$

Q219. Probability of either A or B occurring.

A219. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Q220. A bag has 5 red and 3 blue balls. Probability of drawing red.

A220. $\frac{5}{8}$

Q221. A bag has 4 black and 6 white balls. Probability of drawing white.

A221. $\frac{6}{10} = \frac{3}{5}$

Q222. Number of outcomes when tossing 3 coins.

A222. 8

Q223. Probability of getting 2 heads when tossing 3 coins.

A223. $\frac{3}{8}$

Q224. Number of permutations of 5 distinct objects.

A224. 120

Q225. Formula for permutation.

A225. $P(n, r) = \frac{n!}{(n-r)!}$

Q226. Formula for combination.

A226. $C(n, r) = \frac{n!}{r!(n-r)!}$

Q227. Number of ways to choose 2 from 5.

A227. 10

Q228. Number of ways to arrange 3 out of 5.

A228. 60

Q229. Probability of sum = 7 when rolling 2 dice.

A229. $\frac{6}{36} = \frac{1}{6}$

Q230. Probability of sum = 2 when rolling 2 dice.

A230. $\frac{1}{36}$

Q231. Probability of sum = 12 when rolling 2 dice.

A231. $\frac{1}{36}$

Q232. Mean of 2, 4, 6, 8, 10.

A232. 6

Q233. Median of 12, 18, 15, 10, 20.

A233. 15

Q234. Mode of 2, 2, 3, 4, 5.

A234. 2

Q235. Range of 40, 50, 60, 80.

A235. 40

Q236. If mean = 50 and total of 5 items = ?, what is total?

A236. 250

Q237. Mean of first n natural numbers.

A237. $\frac{n+1}{2}$

Q238. Formula for weighted mean.

A238. $\frac{\sum wx}{\sum w}$

Q239. Quartile that represents the median.

A239. Q2

Q240. Formula for z-score.

A240. $z = \frac{x-\mu}{\sigma}$

Q241. In normal distribution, mean = median = ?

A241. Mode

Q242. Normal curve is also called?

A242. Bell curve

Q243. Approx. % of data within 1σ in normal distribution.

A243. 68%

Q244. Approx. % of data within 2σ.

A244. 95%

Q245. Approx. % of data within 3σ.

A245. 99.7%

Q246. Probability of getting at least one head in 2 tosses.
A246. $\frac{3}{4}$

Q247. Probability of getting exactly 2 heads in 2 tosses.
A247. $\frac{1}{4}$

Q248. Probability of drawing a red card or a black card.
A248. 1

Q249. Probability of drawing a face card from deck.
A249. $\frac{12}{52} = \frac{3}{13}$

Q250. Probability of drawing a spade or a king.
A250. $\frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$

📘 Word Problems (Q251–Q300)

Q251. If a shirt costs ₱350 and is discounted by 20%, what is the sale price?
A251. ₱280

Q252. A car travels 60 km in 1.5 hours. Find speed.
A252. 40 km/h

Q253. If speed = 80 km/h and time = 3 h, distance = ?
A253. 240 km
Q254. If 5 workers finish a job in 6 days, how many days will 10 workers take?
A254. 3 days

Q255. A father is 30 years older than his son. If father is 50, how old is son?
A255. 20

Q256. A number is increased by 20% and becomes 60. Find original.
A256. 50

Q257. Simple interest on ₱5000 at 10% for 2 years.
A257. ₱1000

Q258. Compound interest on ₱5000 at 10% for 2 years.
A258. ₱1050

Q259. If 2x + 3 = 11, find x.
A259. 4

Q260. If 5x – 7 = 18, find x.
A260. 5

Q261. If $x^2 = 49$, find x.
A261. ±7

Q262. If 4x = 20, find x.
A262. 5

Q262. If 4x = 20, find x.
A262. 5

Q263. If 3x + 2 = 14, find x.
A263. 4

Q264. If (x + 3)(x – 3) = ?,
A264. $x^2 - 9$

Q265. Solve: $x^2 - 5x + 6 = 0$.
A265. x = 2 or x = 3

Q266. Solve: $x^2 + 7x + 10 = 0$.
A266. x = –2, –5

Q267. Solve: $2x^2 - 18 = 0$.
A267. x = ±3

Q268. Factor: $x^2 - 16$.
A268. (x – 4)(x + 4)

Q269. Factor: $x^2 + 6x + 9$.
A269. (x + 3)(x + 3)

Q270. Factor: $x^2 - x - 6$.
A270. (x – 3)(x + 2)
Q271. Roots of quadratic equation $ax^2 + bx + c = 0$.
A271. $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Q272. If discriminant = 0, roots are?
A272. Real and equal

Q273. If discriminant > 0, roots are?
A273. Real and unequal

Q274. If discriminant < 0, roots are?
A274. Imaginary / Complex

Q275. Ratio of 2 to 5 expressed as fraction.
A275. $\frac{2}{5}$

Q276. If 3x = 15, ratio of x to 15.
A276. 1 : 5

Q277. A recipe calls for 2 cups sugar to 3 cups flour. Ratio?
A277. 2:3

Q278. If y varies directly as x, and y=10 when x=5, find y when x=8.
A278. 16

Q279. If y varies inversely as x, and y=6 when x=2, find y when x=4.
A279. 3

Q280. Average of 5 numbers is 20. Sum of numbers?
A280. 100

Q281. A student scored 75, 80, 85. What is mean?
A281. 80

Q282. If ₱5000 is shared in ratio 2:3, larger share = ?
A282. ₱3000

Q283. A train 100 m long passes a man in 10 s. Speed?
A283. 10 m/s

Q284. A train 100 m long passes a platform 50 m long in 15 s. Speed?
A284. 10 m/s

Q285. A man walks 5 km east, then 12 km north. Distance from start?
A285. 13 km

Q286. Two numbers have sum 30 and difference 10. Find numbers.
A286. 20 and 10

Q287. Two numbers multiply to 48 and divide to 3. Find numbers.
A287. 12 and 4

Q288. If perimeter of square = 40, find side.
A288. 10

Q289. If area of square = 64, find side.
A289. 8

Q290. If circumference of circle = 31.4, find radius ($\pi = 3.14$).
A290. 5

Q291. If area of circle = 78.5, find radius ($\pi = 3.14$).
A291. 5

Q292. If speed = distance/time, what is time?
A292. distance/speed

Q293. If 8 men finish work in 12 days, how many days for 4 men?
A293. 24 days

Q294. If 3 men eat 6 apples in 2 days, how many for 6 men in 2 days?
A294. 12 apples

Q295. A shirt is ₱600 with 25% discount. Find price.
A295. ₱450

Q296. If 12 mangoes cost ₱96, find cost of 20 mangoes.
A296. ₱160

Q297. Speed of car: 60 km/h. Time to cover 150 km?
A297. 2.5 h

Q298. A tank fills 1/4 in 6 min. Full tank in?
A298. 24 min

Q299. A box has 12 red and 8 blue marbles. Probability of red?
A299. 12/20 = 0.6

Q300. A die is rolled. Probability of odd number?
A300. 3/6 = 1/2

🔢 Algebra & Functions (Q301–Q350)

Q301. If $f(x) = 2x + 3$, find $f(4)$.
A301. 11

Q302. If $f(x) = x^2 - 5$, find $f(3)$.
A302. 4

Q303. If $f(x) = 3x - 7$, find $f(0)$.
A303. –7

Q304. If $f(x) = 2x^2 + 3x + 1$, find $f(-2)$.
A304. 3

Q305. Domain of $f(x) = \sqrt{x}$.
A305. $x \geq 0$

Q306. Domain of $f(x) = \frac{1}{x}$.
A306. $x \neq 0$

Q307. If $f(x) = x + 2$ and $g(x) = x^2$, find $(f \circ g)(2)$.
A307. 6

Q308. If $f(x) = x^2$ and $g(x) = 3x$, find $(g \circ f)(2)$.
A308. 12

Q309. Slope of line through (2,4) and (6,10).
A309. $\frac{10-4}{6-2} = \frac{6}{4} = \frac{3}{2}$

Q310. Equation of line with slope 2 passing through (1,3).
A310. $y - 3 = 2(x - 1)$

Q311. Equation of line through (0,2) slope –3.
A311. $y = -3x + 2$

Q312. Slope of horizontal line.
A312. 0

Q313. Slope of vertical line.
A313. Undefined

Q314. Equation of x-axis.
A314. y = 0

Q315. Equation of y-axis.
A315. x = 0

Q316. Parabola $y = x^2$ opens?
A316. Upward

Q317. Vertex of $y = x^2 + 4x + 5$.
A317. (–2, 1)

Q318. Axis of symmetry of $y = x^2 + 6x + 8$.

A318. x = –3

Q319. Discriminant of $x^2 - 4x + 4$.

A319. 0

Q320. Roots of $x^2 + 4x + 3 = 0$.

A320. –1, –3

Q321. Roots of $x^2 - 9 = 0$.

A321. ±3

Q322. Solve: $3x^2 - 12 = 0$.

A322. ±2

Q323. Sum of roots of $x^2 + 7x + 10 = 0$.

A323. –7

Q324. Product of roots of $x^2 + 7x + 10 = 0$.

A324. 10

Q325. Function that passes vertical line test.

A325. A function

Q326. Inverse of $f(x) = 2x + 3$.

A326. $f^{-1}(x) = \frac{x-3}{2}$

Q327. If slope of a line = m, slope of perpendicular line?

A327. $-\frac{1}{m}$

Q328. If slope of line = 3, slope of perpendicular line = ?

A328. –1/3

Q329. If line is parallel to y = 2x + 5, slope = ?

A329. 2

Q330. Simplify: $(x^2)^3$.

A330. x^6

Q331. Simplify: $x^a \cdot x^b$.

A331. x^{a+b}

Q332. Simplify: $\frac{x^5}{x^2}$.

A332. x^3

Q333. Simplify: $(x^3)^2$.

A333. x^6

Q334. Simplify: x^0.

A334. 1

Q335. Solve: $2^x = 16$.

A335. 4

Q336. Solve: $3^x = 81$.
A336. 4

Q337. Solve: $5^x = 125$.
A337. 3

Q338. Solve: $\log_{10} 1000$.
A338. 3

Q339. Solve: $\log_2 8$.
A339. 3

Q340. Solve: $\log_5 25$.
A340. 2

Q341. Natural logarithm base is?
A341. e (≈ 2.718)

Q342. $\ln e^3 = ?$
A342. 3

Q343. Change of base formula:
A343. $\log_a b = \frac{\log b}{\log a}$

Q344. Exponential growth formula.
A344. $y = y_0 e^{rt}$

Q345. Exponential decay formula.
A345. $y = y_0 e^{-rt}$

Q346. Solve: $x^2 – 25 = 0$.
A346. ±5

Q347. Simplify: $(a + b)^2$.
A347. $a^2 + 2ab + b^2$

Q348. Simplify: $(a - b)^2$.
A348. $a^2 - 2ab + b^2$

Q349. Simplify: $(a + b)(a - b)$.
A349. $a^2 - b^2$

Q350. Sum of arithmetic series formula.
A350. $S_n = \frac{n}{2}(a_1 + a_n)$

📚 Math Trivia & General Knowledge (Q351–Q400)

Q351. Who is the "Father of Geometry"?
A351. Euclid

Q352. Who is the "Prince of Mathematics"?
A352. Carl Friedrich Gauss

Q353. Who invented analytic geometry?
A353. René Descartes

Q354. Who discovered the Pythagorean theorem?
A354. Pythagoras

Q355. Who developed calculus independently of Newton?
A355. Gottfried Wilhelm Leibniz

Q356. Who introduced set theory?
A356. Georg Cantor

Q357. Who is known as the "Father of Algebra"?
A357. Al-Khwarizmi

Q358. The symbol π was popularized by?
A358. William Jones

Q359. Approximate value of π to 3 decimals.
A359. 3.142

Q360. Golden ratio symbol.
A360. ϕ (phi)

Q361. Approximate value of golden ratio.
A361. 1.618

Q362. Who introduced Fibonacci sequence?
A362. Leonardo of Pisa (Fibonacci)

Q363. First five Fibonacci numbers.
A363. 0, 1, 1, 2, 3

Q364. Pascal’s Triangle is named after?
A364. Blaise Pascal

Q365. Who is known for the law of large numbers?
A365. Jacob Bernoulli

Q366. Who created Boolean algebra?
A366. George Boole

Q367. Who proved Fermat’s Last Theorem?
A367. Andrew Wiles

Q368. Who is called the "Human Computer"?
A368. Shakuntala Devi

Q369. Who is known as the "Father of Probability"?
A369. Pierre-Simon Laplace

Q370. The number e ≈ ?
A370. 2.718

Q371. Euler's famous identity.
A371. $e^{i\pi} + 1 = 0$

Q372. Who formulated the laws of motion and gravity?
A372. Isaac Newton

Q373. Which mathematician worked on relativity?
A373. Albert Einstein

Q374. Which Filipino mathematician was first PhD in Math?
A374. Dr. Jose A. Marasigan

Q375. Year the number zero was widely used in Europe.
A375. 12th century

Q376. Who introduced Cartesian coordinate plane?
A376. René Descartes

Q377. What does QED mean in math proofs?
A377. Quod Erat Demonstrandum (which was to be proven)

Q378. Infinity symbol (∞) was introduced by?
A378. John Wallis

Q379. Prime number after 97?
A379. 101

Q380. Largest known prime numbers are found using?
A380. Mersenne primes

Q381. Smallest prime number.
A381. 2

Q382. Is 1 a prime number?
A382. No

Q383. Twin primes are?
A383. Primes differing by 2 (e.g., 11 and 13)

Q384. Perfect number less than 30?
A384. 6 and 28

Q385. Euler's number is symbolized by?
A385. e

Q386. Who created geometry of curved surfaces (non-Euclidean geometry)?
A386. Nikolai Lobachevsky

Q387. Who proved that circle squaring is impossible?
A387. Ferdinand von Lindemann

Q388. The "unsolvable" quintic equation was proven by?
A388. Niels Henrik Abel

Q389. Which mathematician invented graph theory?
A389. Leonhard Euler

Q390. Year decimal system introduced in Europe.
A390. 1202 (Fibonacci's Liber Abaci)

Q391. Who invented logarithms?
A391. John Napier

Q392. Who proved Gödel's incompleteness theorems?
A392. Kurt Gödel

Q393. Who discovered chaos theory?
A393. Henri Poincaré

Q394. Who was nicknamed "Doctor Mirabilis" (Wonderful Teacher)?
A394. Roger Bacon

Q395. The word "algebra" comes from which language?
A395. Arabic

Q396. Who is credited with early work in matrix theory?
A396. Arthur Cayley

Q397. The irrationality of $\sqrt{2}$ was discovered by?
A397. Hippasus of Metapontum

Q398. Who developed group theory?
A398. Évariste Galois

Q399. Who is known as the father of topology?
A399. Henri Poincaré

Q400. Who introduced the mathematical symbol "="?
A400. Robert Recorde

🔢 Advanced Algebra & Functions (Q401–Q450)

Q401. Simplify: $(2x^3)(3x^2)$.
A401. $6x^5$

Q402. Expand: $(x+3)(x+5)$.
A402. $x^2+8x+15$

Q403. Expand: $(x-4)(x+2)$.
A403. x^2-2x-8

Q404. Factor: x^2-9.
A404. $(x-3)(x+3)$

Q405. Factor: x^2+5x+6.
A405. $(x+2)(x+3)$

Q406. Factor: x^2+x-12.
A406. $(x+4)(x-3)$

Q407. Simplify: $\frac{2x^2y}{4xy^2}$.
A407. $\frac{x}{2y}$

Q408. Simplify: $\frac{a^2b^3}{ab^2}$.
A408. ab

Q409. Solve for x: $2x + 5 = 17$.
A409. 6

Q410. Solve for x: $3x - 7 = 11$.
A410. 6

Q411. Solve for x: $\frac{x}{4} + 3 = 5$.
A411. 8

Q412. Solve: $\frac{2x}{3} = 8$.
A412. 12

Q413. Solve: $4(x - 2) = 12$.
A413. 5

Q414. Solve: $\frac{x+1}{2} = 7$.
A414. 13

Q415. Solve: $5x + 2 = 3x + 10$.
A415. 4

Q416. Solve: $2(x + 3) = 14$.
A416. 4

Q417. Solve: $3(x - 5) = 6$.
A417. 7

Q418. Solve: $\frac{x}{5} + 2 = 6$.
A418. 20

Q419. Solve quadratic: $x^2 - 7x + 12 = 0$.
A419. x = 3, 4

Q420. Solve quadratic: $x^2 + 2x - 15 = 0$.
A420. x = 3, –5

Q421. Solve quadratic: $x^2 - 16 = 0$.
A421. x = ±4

Q422. Solve quadratic: $x^2 - 2x - 8 = 0$.
A422. x = 4, –2

Q423. Solve quadratic: $x^2 + 6x + 9 = 0$.
A423. x = –3 (double root)

Q424. Rational root of $x^2 - 5x + 6 = 0$.
A424. 2, 3

Q425. Roots of cubic $x^3 - 6x^2 + 11x - 6 = 0$.
A425. 1, 2, 3

Q426. If sum of roots = –b/a, what is sum of roots of $2x^2 + 3x - 5 = 0$?
A426. –3/2

Q427. Product of roots formula for ax²+bx+c=0.
A427. c/a

Q428. Discriminant formula.
A428. b² – 4ac

Q429. If discriminant > 0, roots are?
A429. Real and distinct

Q430. If discriminant = 0, roots are?
A430. Real and equal

Q431. If discriminant < 0, roots are?
A431. Complex conjugates

Q432. Simplify: $\sqrt{50}$.
A432. $5\sqrt{2}$

Q433. Simplify: $\sqrt{72}$.
A433. $6\sqrt{2}$

Q434. Simplify: $\sqrt{98}$.
A434. $7\sqrt{2}$

Q435. Simplify: $\sqrt{45}$.
A435. $3\sqrt{5}$

Q436. Simplify: $\sqrt{27}$.
A436. $3\sqrt{3}$

Q437. Simplify: $\sqrt{200}$.
A437. $10\sqrt{2}$

Q438. Rationalize: $\frac{1}{\sqrt{3}}$.
A438. $\frac{\sqrt{3}}{3}$

Q439. Rationalize: $\frac{1}{\sqrt{5}}$.
A439. $\frac{\sqrt{5}}{5}$

Q440. Rationalize: $\frac{2}{\sqrt{7}}$.
A440. $\frac{2\sqrt{7}}{7}$

Q441. Solve exponential: $2^x = 32$.
A441. 5

Q442. Solve exponential: $3^x = 243$.
A442. 5

Q443. Solve exponential: $5^x = 625$.
A443. 4

Q444. Simplify: $\log_{10} 10000$.
A444. 4

Q445. Simplify: $\log_2 16$.
A445. 4

Q446. Simplify: $\log_5 125$.
A446. 3

Q447. Simplify: $\ln 1$.
A447. 0

Q448. Simplify: $\ln e$.
A448. 1

Q449. Simplify: $\ln e^5$.
A449. 5

Q450. Solve: $\log_3 9 = ?$
A450. 2

🎲 Probability, Statistics & Trivia (Q451–Q500)

Q451. Probability of tossing heads in one coin flip.
A451. 1/2

Q452. Probability of rolling a 6 on one die.
A452. 1/6
Q453. Probability of rolling an even number on a die.
A453. 3/6 = 1/2

Q454. Probability of rolling number > 4 on a die.
A454. 2/6 = 1/3

Q455. Probability of drawing an Ace from a 52-card deck.
A455. 4/52 = 1/13

Q456. Probability of drawing a Heart card.
A456. 13/52 = 1/4

Q457. Probability of drawing a King of Hearts.
A457. 1/52

Q458. Probability of drawing a red card.
A458. 26/52 = 1/2

Q459. Probability of drawing a face card (J, Q, K).
A459. 12/52 = 3/13

Q460. Probability of drawing a spade or club.
A460. 26/52 = 1/2

Q461. If two coins tossed, probability of 2 heads.
A461. 1/4

Q462. If two dice rolled, probability of sum = 7.
A462. 6/36 = 1/6

Q463. Mean of 2, 4, 6, 8.
A463. 5

Q464. Median of 2, 4, 6, 8.
A464. 5

Q465. Mode of 2, 4, 6, 8.
A465. None

Q466. Range of 2, 4, 6, 8.
A466. 6

Q467. Variance of 2, 4, 6, 8.
A467. 5

Q468. Standard deviation of 2, 4, 6, 8.
A468. $\sqrt{5}$

Q469. Probability of drawing 2 aces from a 52-card deck.
A469. $\frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}$

Q470. Number of ways to arrange 4 objects.
A470. 4! = 24

Q471. Number of ways to arrange 5 books.
A471. 5! = 120

Q472. Combination formula.
A472. $C(n, r) = \frac{n!}{r!(n-r)!}$

Q473. Permutation formula.
A473. $P(n, r) = \frac{n!}{(n-r)!}$

Q474. Number of ways to choose 2 from 5.
A474. 10

Q475. Number of ways to arrange 2 from 5.
A475. 20

Q476. Probability that leap year has 53 Sundays.
A476. 2/7

Q477. Birthday paradox: probability ≥50% with how many people?
A477. 23

Q478. Law of Large Numbers is attributed to?
A478. Jacob Bernoulli

Q479. Bell-shaped curve is known as?
A479. Normal distribution

Q480. Area under normal curve equals?
A480. 1

Q481. Mean, median, mode are equal in what distribution?
A481. Normal distribution

Q482. Z-score formula.
A482. $z = \frac{x-\mu}{\sigma}$

Q483. Probability of rolling prime number on die.
A483. 3/6 = 1/2

Q484. Probability of rolling multiple of 3 on die.
A484. 2/6 = 1/3

Q485. Probability of drawing queen from 52 cards.
A485. 4/52 = 1/13

Q486. Probability of drawing black king.
A486. 2/52 = 1/26

Q487. Probability of selecting vowel from ENTRANCE.
A487. 3/8

Q488. Probability of vowel from word "PROBABILITY".
A488. 4/11

Q489. Probability of consonant from "STATISTICS".
A489. 6/10 = 3/5

Q490. Probability of vowel from "ALGEBRA".
A490. 3/7

Q491. Who created probability theory?
A491. Blaise Pascal and Pierre de Fermat

Q492. Who is known for statistics in social sciences?
A492. Francis Galton

Q493. Karl Pearson is known for what test?
A493. Chi-square test

Q494. Ronald Fisher is known for?
A494. Analysis of Variance (ANOVA)

Q495. What does σ^2 represent?
A495. Variance

Q496. What does σ represent?
A496. Standard deviation

Q497. What does μ represent in statistics?
A497. Population mean

Q498. What does $\bar{x}$ represent in statistics?
A498. Sample mean

Q499. Probability of rolling sum of 11 with two dice.
A499. 2/36 = 1/18

Q500. Probability of rolling sum of 2 with two dice.
A500. 1/36

📐 Geometry (Q501–Q550)

Q501. Sum of interior angles of a triangle.
A501. 180°

Q502. Sum of interior angles of a quadrilateral.
A502. 360°

Q503. Sum of interior angles of a pentagon.
A503. 540°

Q504. Sum of interior angles of a hexagon.
A504. 720°

Q505. Formula for sum of interior angles of n-gon.
A505. $(n-2) \times 180^\circ$

Q506. Each interior angle of a regular hexagon.
A506. 120°

Q507. Each interior angle of a regular pentagon.
A507. 108°

Q508. Each interior angle of a regular octagon.
A508. 135°
Q509. Formula for exterior angle of a regular polygon.
A509. $360^\circ / n$

Q510. Number of diagonals in a pentagon.
A510. 5

Q511. Formula for diagonals of n-gon.
A511. $\frac{n(n-3)}{2}$

Q512. Area of triangle formula.
A512. $\frac{1}{2}bh$

Q513. Heron's formula.
A513. $A = \sqrt{s(s-a)(s-b)(s-c)}$, where $s = \frac{a+b+c}{2}$

Q514. Area of circle.
A514. πr^2

Q515. Circumference of circle.
A515. $2\pi r$

Q516. Area of square.
A516. s^2

Q517. Area of rectangle.
A517. lw

Q518. Area of trapezoid.
A518. $\frac{1}{2}(b_1 + b_2)h$

Q519. Area of parallelogram.
A519. bh

Q520. Volume of cube.
A520. s^3

Q521. Volume of rectangular prism.
A521. lwh

Q522. Volume of cylinder.
A522. $\pi r^2 h$

Q523. Volume of sphere.
A523. $\frac{4}{3}\pi r^3$

Q524. Volume of cone.
A524. $\frac{1}{3}\pi r^2 h$

Q525. Pythagorean theorem.
A525. $a^2 + b^2 = c^2$

Q526. Longest side in a right triangle.
A526. Hypotenuse

Q527. Distance formula.
A527. $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Q528. Midpoint formula.
A528. $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

Q529. Equation of circle (center at origin, radius r).
A529. $x^2 + y^2 = r^2$

Q530. Equation of circle (center (h,k), radius r).
A530. $(x - h)^2 + (y - k)^2 = r^2$

Q531. Surface area of sphere.
A531. $4\pi r^2$

Q532. Surface area of cube.
A532. $6s^2$

Q533. Surface area of cylinder.
A533. $2\pi r^2 + 2\pi r h$

Q534. Ratio of circumference to diameter of a circle.
A534. π

Q535. Diagonal of square formula.
A535. $s\sqrt{2}$

Q536. Diagonal of rectangle formula.
A536. $\sqrt{l^2 + w^2}$

Q537. Height of equilateral triangle with side s.
A537. $\frac{\sqrt{3}}{2}s$

Q538. Area of equilateral triangle.
A538. $\frac{\sqrt{3}}{4}s^2$

Q539. Formula for area of rhombus (diagonals d_1, d_2).
A539. $\frac{1}{2}d_1d_2$

Q540. A polygon with 12 sides.
A540. Dodecagon

Q541. Polygon with 20 sides.
A541. Icosagon

Q542. A polygon with 100 sides.
A542. Hectogon

Q543. Polyhedron with 6 faces.
A543. Cube (or hexahedron)

Q544. Solid with all faces triangles.
A544. Tetrahedron

Q545. How many faces does an octahedron have?
A545. 8

Q546. How many faces does an icosahedron have?
A546. 20

Q547. Euler's polyhedron formula.
A547. $V - E + F = 2$

Q548. A 3D figure with circular base and pointy top.
A548. Cone

Q549. A 3D figure with two circular bases.
A549. Cylinder

Q550. A solid with no edges or vertices.
A550. Sphere

📐 Trigonometry (Q551–Q580)

Q551. sin 0° = ?
A551. 0

Q552. cos 0° = ?
A552. 1

Q553. tan 0° = ?
A553. 0

Q554. sin 30° = ?
A554. 1/2

Q555. cos 30° = ?
A555. $\sqrt{3}/2$

Q556. tan 30° = ?
A556. $1/\sqrt{3}$

Q557. sin 45° = ?
A557. $\sqrt{2}/2$

Q558. cos 45° = ?
A558. $\sqrt{2}/2$

Q559. tan 45° = ?
A559. 1

Q560. sin 60° = ?
A560. $\sqrt{3}/2$

Q561. cos 60° = ?
A561. 1/2
Q562. tan 60° = ?
A562. $\sqrt{3}$

Q563. sin 90° = ?
A563. 1

Q564. cos 90° = ?
A564. 0

Q565. tan 90° = ?
A565. Undefined

Q566. Pythagorean identity.
A566. $\sin^2\theta + \cos^2\theta = 1$

Q567. tan θ = ? (in terms of sin and cos)
A567. $\frac{\sin\theta}{\cos\theta}$

Q568. cot θ = ?
A568. $\frac{\cos\theta}{\sin\theta}$

Q569. sec θ = ?
A569. $\frac{1}{\cos\theta}$

Q570. csc θ = ?
A570. $\frac{1}{\sin\theta}$

Q571. Law of Sines formula.
A571. $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Q572. Law of Cosines formula.
A572. $c^2 = a^2 + b^2 - 2ab\cos C$

Q573. sin(–θ) = ?
A573. –sin θ

Q574. cos(–θ) = ?
A574. cos θ

Q575. tan(–θ) = ?
A575. –tan θ

Q576. Angle sum identity: sin(A+B).
A576. $\sin A\cos B + \cos A\sin B$

Q577. cos(A+B).
A577. $\cos A\cos B - \sin A\sin B$

Q578. tan(A+B).
A578. $\frac{\tan A + \tan B}{1 - \tan A\tan B}$

Q579. Double-angle: sin 2θ.
A579. 2 sinθ cosθ

Q580. Double-angle: cos 2θ.
A580. cos²θ – sin²θ

🧩 Math Trivia (Q581–Q600)

Q581. Who is the “Father of Trigonometry”?
A581. Hipparchus

Q582. Who introduced the term “sine”?
A582. Aryabhata

Q583. Who invented coordinate geometry?
A583. René Descartes

Q584. What does radian measure?
A584. Angle based on arc length/radius

Q585. How many radians in a circle?
A585. 2π radians

Q586. Degrees equivalent to 1 radian.
A586. ≈ 57.3°

Q587. Who is known as the “Father of Calculus”?
A587. Isaac Newton & Gottfried Leibniz

Q588. Who is the “Queen of Mathematics”?
A588. Number theory

Q589. The Collatz conjecture is also called?
A589. 3n+1 problem

Q590. Who is the first woman mathematician in history?
A590. Hypatia of Alexandria

Q591. Who solved Fermat’s Last Theorem?
A591. Andrew Wiles

Q592. Who discovered π to high precision using polygons?
A592. Archimedes

Q593. Who first used negative numbers in China?
A593. Liu Hui

Q594. The word “geometry” means?
A594. Earth measurement

Q595. What’s the shape with minimum perimeter for given area?
A595. Circle

Q596. What’s the flattest 3D shape (2D object)?
A596. Plane

Q597. Who invented Pascal’s Triangle before Pascal?
A597. Chinese mathematicians (Yang Hui)

Q598. Who introduced infinity symbol ∞?
A598. John Wallis

Q599. What does i represent in mathematics?
A599. Imaginary unit ($\sqrt{-1}$)

Q600. What is Euler’s identity?
A600. $e^{i\pi} + 1 = 0$

📐 Advanced Trigonometry (Q601–Q640)

Q601. sin²θ = ? (in terms of cos 2θ)
A601. $\frac{1-\cos 2\theta}{2}$

Q602. cos²θ = ? (in terms of cos 2θ)
A602. $\frac{1+\cos 2\theta}{2}$

Q603. tan²θ = ? (in terms of sec²θ)
A603. $\sec^2\theta - 1$

Q604. sec²θ = ? (in terms of tan²θ)
A604. $1 + \tan^2\theta$

Q605. csc²θ = ? (in terms of cot²θ)
A605. $1 + \cot^2\theta$

Q606. sin(A – B).
A606. $\sin A\cos B - \cos A\sin B$

Q607. cos(A – B).
A607. $\cos A\cos B + \sin A\sin B$

Q608. tan(A – B).
A608. $\frac{\tan A - \tan B}{1+\tan A\tan B}$

Q609. Half-angle identity: sin²(θ/2).

A609. $\frac{1-\cos\theta}{2}$

Q610. Half-angle identity: cos²(θ/2).

A610. $\frac{1+\cos\theta}{2}$

Q611. Double-angle: tan 2θ.

A611. $\frac{2\tan\theta}{1-\tan^2\theta}$

Q612. Product-to-sum: sin A sin B.

A612. $\frac{1}{2}[\cos(A-B)-\cos(A+B)]$

Q613. Product-to-sum: cos A cos B.

A613. $\frac{1}{2}[\cos(A-B)+\cos(A+B)]$

Q614. Product-to-sum: sin A cos B.

A614. $\frac{1}{2}[\sin(A+B)+\sin(A-B)]$

Q615. Angle of elevation.

A615. Angle formed above the horizontal line of sight.

Q616. Angle of depression.

A616. Angle formed below the horizontal line of sight.

Q617. Inverse of sin θ.

A617. arcsin θ or $\sin^{-1}\theta$